

#AI/ML Training Syllabus

4-Month Training (16 Weeks)

- Duration: 4 Months (16 Weeks)
- Class Timing: 2 Hours per Day
- Assessments: Every Friday (Weekly Assessment)



Catalog

4-Month Artificial Intelligence & Machine Learning Training Journey



01

1. Course Overview

03

3. Month 2 (Weeks 5–8): Machine Learning Core

05

5. Month 4 (Weeks 13–16): Project & Career Preparation

02

2. Month 1 (Weeks 1–4): Python, Mathematics & AI Fundamentals

04

4. Month 3 (Weeks 9–12): Deep Learning & Advanced AI

06

6. Deliverables & Final Evaluation

Course Overview



DURATION

4 Months (16 Weeks)



CLASS TIMING

2 Hours per Day



ASSESSMENTS

Every Friday (Weekly Assessment)

TOOLS & TECH

Python

NumPy

Pandas

Scikit-learn

TensorFlow/Keras

NLP Basics

OpenCV (Intro)

SQL

Git



Outcome: Strong AI & Machine Learning foundation with Deep Learning project

Month 1: Python, Mathematics & AI Fundamentals

W1

Week 1: Introduction to Artificial Intelligence & Python Basics

- > What is Artificial Intelligence?
- > AI vs ML vs Deep Learning
- > Applications of AI
- > AI lifecycle
- > Python basics

W2

Week 2: Python Programming for AI

- > Conditional statements
- > Loops
- > Functions
- > Exception handling
- > File handling
- > Git basics

Month 1: Mathematics & Data Handling

W3

Week 3: Mathematics for Machine Learning

- √x Linear Algebra basics
- 📊 Matrix operations
- ∞ Basic calculus
- 🎲 Probability fundamentals
- 🧠 Bayes theorem

W4

Week 4: Data Handling with Python

- 📦 NumPy arrays
- 📄 Pandas DataFrames
- 🕒 Data visualization basics

KEY FOCUS

Building the analytical foundation for Month 2

Month 2: Machine Learning Core

CONCEPTS

05

Week 5: Introduction to Machine Learning

- Supervised, Unsupervised, Reinforcement Learning
- Bias & Variance
- ML workflow

ALGORITHMS

06

Week 6: Regression Algorithms

- Linear Regression
- Evaluation metrics
- Polynomial Regression

Month 2: Classification & Unsupervised Learning

07 Week 7: Classification Algorithms

- Logistic Regression
- KNN
- Decision Trees
- Confusion Matrix

08 Week 8: Unsupervised Learning

- K-Means
- Hierarchical clustering
- PCA

 CORE ML MODULE 02

Month 3: Deep Learning & Advanced AI

09

Week 9: Neural Network Fundamentals

- Perceptron
- Activation functions
- Backpropagation

10

Week 10: Artificial Neural Networks

- Building ANN using TensorFlow/Keras
- Optimizers
- Regularization

Month 3: Computer Vision & NLP

11

📺 Week 11: Computer Vision

- ✓ OpenCV basics
- ✓ CNN concept

12

📖 Week 12: Natural Language Processing

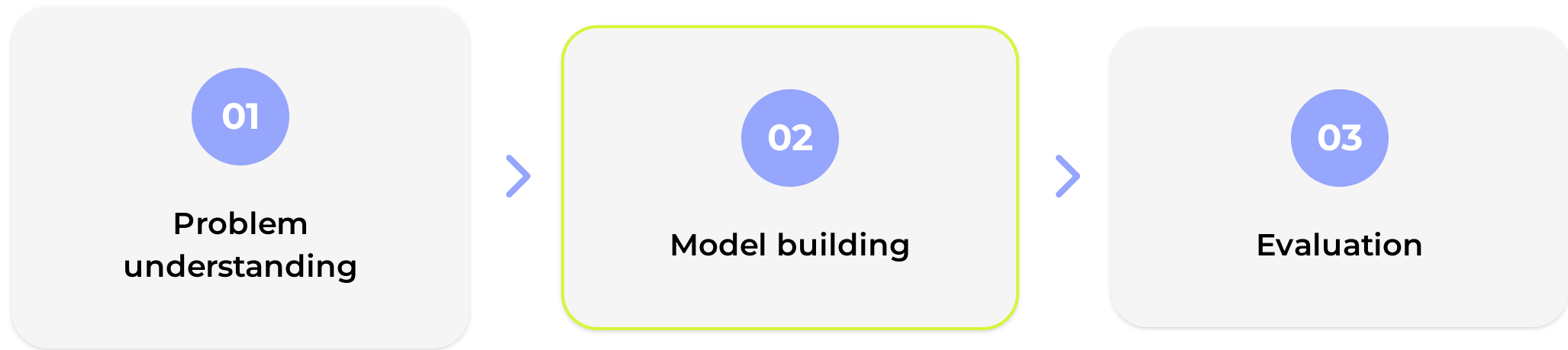
- ✓ Text preprocessing
- ✓ TF-IDF
- ✓ Sentiment analysis



Advanced AI modules focusing on unstructured data processing

Month 4 — Capstone Project Implementation

Week 13-14: Capstone Project Implementation

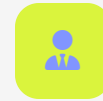


Month 4 — Optimization, Deployment & Career Prep



Week 15: Model Optimization & Deployment Basics

- Hyperparameter tuning
- Model saving/loading
- Intro to Streamlit/Flask



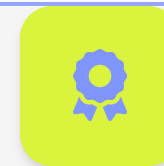
Week 16: Presentation & Interview Preparation

- Resume building
- AI/ML interview questions
- Mock interviews

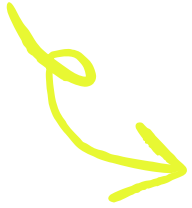
Deliverables & Final Evaluation

Student Deliverables

- ✓ Weekly assessments
- ✓ Python notebooks
- ✓ AI/ML Capstone Project
- ✓ GitHub repository (optional)



Final Evaluation
Capstone Project + Viva
CERTIFICATION REQUIREMENT



Thank You

4-Month AI/ML Training Syllabus

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Page 13

